Practical No. 8: Implement program to use k-Nearest Neighbors (KNN) model for Classification on given dataset.

Q.1) Build KNN Classification model for a given dataset. Vary the number of k values as follows and compare the results:

i. 1 ii. 3 iii. 5

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

```
data = {
    "Temp":[35,38,40,42,45,47,50,52,55,58],
    "Vibration":[1.1,1.5,2.0,2.4,2.8,3.1,3.5,3.8,4.2,4.6],
    "Pressure":[30,32,35,37,40,42,45,47,50,52],
    "Voltage":[220,218,215,213,210,208,205,203,200,198],
    "Current":[5.0,5.5,6.0,6.5,7.0,7.5,8.0,8.5,9.0,9.5],
    "Humidity":[40,42,45,47,50,52,55,57,60,62],
    "Noise":[60,62,65,67,70,72,75,77,80,82],
    "RPM":[1500,1550,1600,1650,1700,1750,1800,1850,1900,1950],
    "Torque":[10,11,12,13,14,15,16,17,18,19],
    "Load":[70,72,75,77,80,82,85,87,90,92],
    "Energy":[100,105,110,115,120,125,130,135,140,145],
    "OilLevel":[80,78,76,74,72,70,68,66,64,62],
    "Wear":[1,1,2,2,3,3,4,4,5,5],
    "Age":[1,1,2,2,3,3,4,4,5,5],
    "Condition":[0,0,0,0,0,1,1,1,1,1]
}

df = pd.DataFrame(data)
print(df)
```

```
   Temp  Vibration  Pressure  Voltage  Current  Humidity  Noise   RPM  Torque  \
0    35        1.1        30      220      5.0        40     60  1500      10
1    38        1.5        32      218      5.5        42     62  1550      11
2    40        2.0        35      215      6.0        45     65  1600      12
3    42        2.4        37      213      6.5        47     67  1650      13
4    45        2.8        40      210      7.0        50     70  1700      14
5    47        3.1        42      208      7.5        52     72  1750      15
6    50        3.5        45      205      8.0        55     75  1800      16
7    52        3.8        47      203      8.5        57     77  1850      17
8    55        4.2        50      200      9.0        60     80  1900      18
9    58        4.6        52      198      9.5        62     82  1950      19

   Load  Energy  OilLevel  Wear  Age  Condition
0    70     100        80     1    1          0
1    72     105        78     1    1          0
2    75     110        76     2    2          0
3    77     115        74     2    2          0
4    80     120        72     3    3          0
5    82     125        70     3    3          1
6    85     130        68     4    4          1
7    87     135        66     4    4          1
8    90     140        64     5    5          1
9    92     145        62     5    5          1
```

```
X = df.drop("Condition",axis=1)
y = df["Condition"]
```

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2, random_state=42)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
knn1 = KNeighborsClassifier(n_neighbors=1)
knn1.fit(X_train,y_train)

pred1 = knn1.predict(X_test)
print(pred1)
```

```
[1 0]
```

```
knn1.score(X_test,y_test)
```

```
1.0
```

```
knn3 = KNeighborsClassifier(n_neighbors=3)
knn3.fit(X_train,y_train)
pred3 = knn3.predict(X_test)
print(pred3)
```

```
[1 0]
```

```
knn3.score(X_test,y_test)
```

```
1.0
```

```
knn5 = KNeighborsClassifier(n_neighbors=5)
knn5.fit(X_train,y_train)
pred5 = knn5.predict(X_test)
pred5
```

```
array([1, 0])
```

```
knn5.score(X_test,y_test)
```

```
1.0
```